Student t-Distribution:

Previously in introductory statistics, we learned about the close relationship between the t and Normal distribution, where t actually became the Normal distribution as $n \to \infty$. This should somewhat make sense reflecting on introductory statistics, since one of the distinguishing differences was the spread of the distributions and how the sample standard deviation and the t distribution were dependent on the degrees of freedom, which was a function of the sample size. $df = (n-1)$

Let $W \sim N(0,1)$ and $V \sim \mathrm{X}^2(r)$, where W, V are independent, then the joint distribution of W and V is:

$$h(w,v) = f(w) \cdot g(v)$$

$$= \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}w^2}\right) \cdot \left(\frac{1}{\Gamma(\frac{r}{2}) 2^{r/2}} v^{\frac{r}{2}-1} e^{-\frac{v}{2}}\right)$$

where $-\infty < w < \infty$
$0 < v < \infty$

Now, let $T = \dfrac{W}{\sqrt{V/r}}$

then through a transformation procedure similar to what we've done before, the PDF for T, a new random variable, can be defined as:

$$f(t) = \frac{\Gamma\left(\frac{r+1}{2}\right)}{\sqrt{\pi \cdot r} \cdot \Gamma\left(\frac{r}{2}\right)} \left(1 + \frac{t^2}{r}\right)^{-\frac{r+1}{2}}$$

If it was not apparent, the random variable T relates to the same type of **statistics** talked about in Definition 4.1.2, i.e. ones based on a random sample. As such, it should not be surprising that T follows the t-distribution that is dependent on r, the degrees of freedom of the X^2-distribution, hence why there is such an important emphasis on the data coming from a normal distribution in introductory statistics since we know the close relationship between the X^2 and normal distribution.

Similar to some other distributions, calculating the kth moment is more efficient than doing the individual moments,

$$E(T^k) = E(W^k) \cdot \frac{2^{-\frac{k}{2}} \Gamma\left(\frac{r}{2} - \frac{k}{2}\right)}{\Gamma\left(\frac{r}{2}\right) r^{-\frac{k}{2}}} \quad \forall\, k < r$$

Thus, the first moment is:

$$E(T) = E(W) \cdot \frac{2^{-\frac{1}{2}}\,\Gamma\left(\frac{r}{2} - \frac{1}{2}\right)}{\Gamma\left(\frac{r}{2}\right)\, r^{-\frac{1}{2}}}$$

Since $W \sim N(0,1)$ then $E(W) = 0$ and therefore $E(T) = 0$. Calculating the second moment to get the variance follows a similar step:

$$\begin{aligned}
Var(T) &= E(T^2) - E(T)^2 \quad (\text{since } E(T)=0) \\
&= E(T^2) \\
&= E(W^2) \cdot \frac{2^{-1}\Gamma\left(\frac{r}{2}-1\right)}{\Gamma\left(\frac{r}{2}\right) r^{-1}} = E(W^2) \cdot \frac{\Gamma\left(\frac{r}{2}-1\right)}{\Gamma\left(\frac{r}{2}\right)}\left(\frac{r}{2}\right) \\
&= E(W^2) \cdot \frac{\left(\frac{r}{2}-2\right)!}{\left(\frac{r}{2}-1\right)!} \cdot \left(\frac{r}{2}\right) \\
&= E(W^2) \cdot \frac{1}{\left(\frac{r}{2}-1\right)} \cdot \left(\frac{r}{2}\right) \\
&= \frac{r}{r-2}
\end{aligned}$$

$\Longrightarrow$ t-distribution w $r > 2 \Rightarrow E(T) = \mu = 0$

$Var(T) = \sigma^2 = \frac{r}{r-2}$

Although not established yet, we know that the Z is closely related to the t-distribution, and here we see that the $\mathrm{Var}(T)$ is a function of $r/(r-2)$, where r is the degrees of freedom, which dependent on the sample size n.

Notice what happens, though, as our sample size n increases. Keeping in mind that the degrees of freedom $r = n - 1$, which we previously defined in introductory statistics.

$$\begin{aligned}
\text{as } n \uparrow &\Longrightarrow df = r \uparrow \\
&\Rightarrow \lim_{n\to\infty} \frac{r}{r-2} = \lim_{r\to\infty} \frac{r}{r-2} \\
&= \lim_{r\to\infty} Var(T) = \sigma^2 = 1
\end{aligned}$$

Theorem 3.6.1: Let $X_1, X_2, \dots, X_n \overset{iid}{\sim} N(\mu, \sigma)$ and $\bar{X} = \frac{1}{n}\sum X_i$ and $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$, then

(A) $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$

Informally, we use corollary 3.4.1 to prove this idea by taking advantage of the properties of the expected value.

More formally, we need thm 3.4.2 where $X_i \sim N(\mu, \sigma)$, $Y = \frac{1}{n}\sum X_i = \bar{X}$, and $a_i = \frac{1}{n}$, then

$$
\begin{aligned}
M_Y(t) = E(e^{tY}) &= E\left(e^{t[\frac{1}{n}\sum X_i]}\right) \\
&= E\left(e^{t\frac{1}{n}X_1 + \dots + t\frac{1}{n}X_n}\right) \\
&= E\left(e^{t\frac{1}{n}X_1} \cdot \dots \cdot e^{t\frac{1}{n}X_n}\right) \\
&= E\left(e^{t\frac{1}{n}X_1}\right) \cdot \dots \cdot E\left(e^{t\frac{1}{n}X_n}\right) \\
&= \prod_{i=1}^{n} E\left(e^{t\frac{1}{n}X_i}\right) \\
&= \prod_{i=1}^{n} \int e^{t\frac{1}{n}x_i} \cdot f(x_i)\, dx_i \\
&= \prod_{i=1}^{n} \int e^{(\frac{1}{n}t)x_i} f(x_i)\, dx_i \\
&= \prod_{i=1}^{n} e^{\mu(\frac{1}{n}t) + \frac{1}{2}\sigma^2(\frac{1}{n}t)^2} \\
&= \prod_{i=1}^{n} e^{\frac{\mu}{n}t + \frac{1}{2}\frac{\sigma^2}{n^2}t^2} \\
&= e^{n(\frac{\mu}{n})t + \frac{1}{2}n(\frac{\sigma^2}{n^2})t^2} \\
&= e^{\mu t + \frac{1}{2}\frac{\sigma^2}{n}t^2} \\
&\Longrightarrow Y \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)
\end{aligned}
$$

(B) $\bar{X}$ and S^2 are independent

(C) The random variable

$$\frac{(n-1)\,S^2}{\sigma^2} \sim \mathrm{X}^2(n-1)$$

(D) The random variable

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

This new random variable T that we are defining is known as a **pivot variable**, which is a random variable that is a function of observations and unobservable parameters

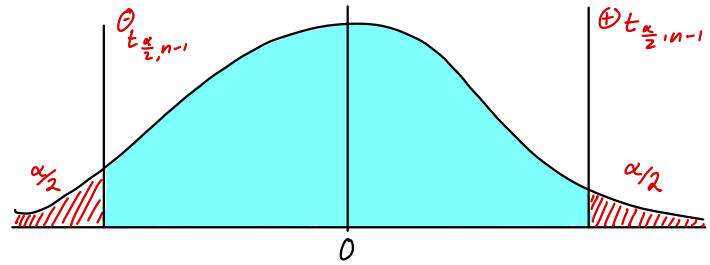
Example 4.5: Suppose $X_1, X_2, \dots, X_n$ is a sample such that $X_i \sim N(\mu, \sigma)$ and $\bar{x}$ and s^2 denote the sample mean and variance. We previously established that

$\bar{X}$ is the MLE(θ) for μ

$S^2\left(\frac{n-1}{n}\right)$ is the MLE(θ) for σ^2

Based on Theorem 3.6.1 (D), we know that

$$T = \frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$$



and if we let $\exists\ \alpha \ni t_{\alpha/2,n-1}$ be the upper $\alpha/2$ tail of the t-distribution: $P\left(T > t_{\frac{\alpha}{2},n-1}\right) = \frac{\alpha}{2}$

$$P\left(-t_{\frac{\alpha}{2},n-1} < T < t_{\frac{\alpha}{2},n-1}\right) = 1-\alpha$$

$$\Leftrightarrow P\left(-t_{\frac{\alpha}{2},n-1} < \frac{\bar{X}-\mu}{S/\sqrt{n}} < t_{\frac{\alpha}{2},n-1}\right) = 1-\alpha$$

$$\Leftrightarrow P\left(\bar{X} - t_{\frac{\alpha}{2},n-1}\cdot\frac{S}{\sqrt{n}} < \mu < \bar{X} + t_{\frac{\alpha}{2},n-1}\cdot\frac{S}{\sqrt{n}}\right) = 1-\alpha$$

Then, once the sample is drawn, there is a $100(1-\alpha)\%$ chance that μ falls within the CI

$$\bar{X} \pm t_{\frac{\alpha}{2},n-1}\cdot\frac{S}{\sqrt{n}}$$

classic formula for the $100(1-\alpha)\%$ CI for μ

margin of error

RECALL from 231, every CI that involves t/z follows the structure est $\pm$ ME $\begin{cases} t_{\frac{\alpha}{2},n-1}\cdot\frac{S}{\sqrt{n}} \\ z_{\frac{\alpha}{2}}\cdot\frac{\sigma}{\sqrt{n}} \end{cases}$

Theorem 4.2.1: Let $X_1, X_2, \ldots, X_n \sim RV$ with some finite μ, σ^2, then the distribution function of the random variable W, where

$$W = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}}$$

converges to $N(0,1)$ as $n \to \infty$.

As we learned in previous examples, although S^2 is the MLE of σ^2, it is **biased**, as the sample size increases, it becomes **asymptotically unbiased**.

$$\lim_{n\to\infty} E(S^2) = \lim_{n\to\infty}\left(\frac{n-1}{n}\right)\sigma^2 = \sigma^2$$

Thus, for some random variable Z_n, where we replace σ with s,

$$\lim_{n\to\infty} Z_n = \lim_{n\to\infty} \frac{\bar{X}-\mu}{s/\sqrt{n}} \sim N(0,1)$$

Using Theorem 4.2.1, if we have $X_1, X_2, \dots, X_n \overset{iid}{\sim} f(x) \ni \exists\, \mu, \sigma^2 < \infty$, then although $f(x)$ **does not have to represent the PDF of the Normal distribution**, we know that Z_n is approximately normal, and therefore we can still calculate an interval similar to the t-based CI.

$$P\left(-Z_{\frac{\alpha}{2}} < \frac{\bar{X}-\mu}{s/\sqrt{n}} < Z_{\frac{\alpha}{2}}\right) \approx 1-\alpha$$

$$\Leftrightarrow P\left(\bar{X} - Z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}} < \mu < \bar{X} + Z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}\right) \approx 1-\alpha$$

and similar to before, a $100(1-\alpha)\%$ CI for μ is

$$X \pm Z_{\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}$$

Not surprisingly, this is often referred to as the **large sample CI for $\boldsymbol{\mu}$**, since for any α, the t-value will create a different size CI and is often associated with small sample sizes and when σ is unknown.

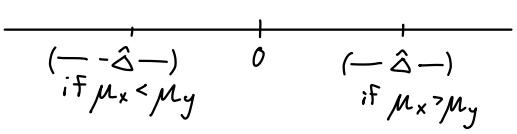
Continuing with the large sample theme, when comparing two population means for X and Y, the CI for the difference in means, $\Delta = \mu_X - \mu_Y$, the CI for Δ can be calculated in a similar fashion.

Question: It is important to note that the order of the difference has no significant impact on the CI and will only cause a shift in sign. Why?

$$EST \pm ME$$

$$est \pm Z_{\frac{\alpha}{2}} \cdot SE(est)$$

$$\hat{\Delta} \pm Z_{\frac{\alpha}{2}} \cdot SE(\hat{\Delta})$$



If we assume equal variance between the two populations, $\sigma_X^2 = \sigma_Y^2$, then if $x_1, x_2, \dots, x_n \sim f(x)$ and $y_1, y_2, \dots, y_m \sim g(y)$, where the n and m samples are **independent** and $\widehat{\Delta} = \bar{x} - \bar{y}$, where $\widehat{\Delta}$ is an unbiased estimator of Δ.

if a statistic is biased $\Rightarrow E(\hat{\theta}) \neq \theta$

$\Rightarrow E(\hat{\Delta}) = E(\bar{x} - \bar{y}) = E(\bar{x}) + E(-\bar{y}) = \mu_x + (-\mu_y) = \mu_x - \mu_y = \Delta$

$\therefore$ $\hat{\Delta}$ is an unbiased estimator

$Var(\hat{\Delta}) = Var(\bar{x} - \bar{y}) = Var(\bar{x}) + Var(-\bar{y}) = Var(\bar{x}) + Var(\bar{y}) = \frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}$

$\hookrightarrow Var\left(\frac{1}{n}\sum x_i\right)$

$= \left(\frac{1}{n}\right)^2 Var\left(\sum x_i\right)$

$\vdots$

$= \frac{\sigma_x^2}{n}$

Let s_x^2, s_y^2 represent the sample variances, then based on the CLT, a $100(1-\alpha)\%$ CI for Δ is:

$est \pm ME$

$\hat{\Delta} \pm Z_{\frac{\alpha}{2}} \cdot SE(\hat{\Delta})$

$= \hat{\Delta} \pm Z_{\frac{\alpha}{2}} \cdot \sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}$

where $Z = \frac{\hat{\Delta} - \Delta}{\sqrt{\frac{S_x^2}{n} + \frac{S_y^2}{m}}} \sim N(0,1)$

It is worth noting that the random variable Z is also a pivot variable, since we can see that it is a function of observations (i.e. the sample means) and unobservable parameters (σ_X^2, σ_Y^2) that we approximate via the sample variances.

In the case of a small sample size and unequal variance, then:

$t = \frac{\hat{\Delta} - \Delta}{\sqrt{\frac{S_p^2}{n} + \frac{S_p^2}{m}}}$

This should look familiar, as it holds a similar structure as the Z-version we just calculated but with one major difference, we use s_p^2 instead of the individual sample variances.

$$s_p^2 = \frac{(n-1)s_x^2 + (m-1)s_y^2}{(n-1)+(m-1)} = \frac{df_x\, s_x^2 + df_y\, s_y^2}{df_x + df_y} = \frac{df_x}{df_x + df_y} s_x^2 + \frac{df_y}{df_x + df_y} s_y^2 \Bigg\} \text{weighted average}$$

but if $n=m$:

$$s_p^2 = \frac{1}{2} s_x^2 + \frac{1}{2} s_y^2 \Longrightarrow \sqrt{\frac{s_p^2}{n} + \frac{s_p^2}{m}} = \sqrt{\frac{s_x^2}{n} + \frac{s_y^2}{m}}$$

Previously, we established that

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1)$$

and via Theorem 3.4.1,

$$Z^2 = \left(\frac{X-\mu}{\sigma}\right)^2 \sim \mathrm{X}^2(1)$$

and via Corollary 3.3.1,

$$\sum Z_i^2 \sim \mathrm{X}^2(r)$$

In Theorem 3.6.1, if $X_i \overset{iid}{\sim} N(\mu, \sigma)$,

(A) $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$, which we proved more formally previously using Theorem 3.4.2.

(B) $\bar{X}, S^2$ are independent

The proof is beyond the scope of the course, but if we let $Y = [X_1 - \bar{X}, X_2 - \bar{X}, \dots, X_n - \bar{X}\,]'$, we can prove that $\mathrm{Cov}(\bar{X}, Y) = 0$, which implies that $\bar{X}, Y$ are independent. Since S^2 is a linear transformation of Y, where $S^2 = \frac{1}{n-1} Y'Y$, then $\bar{X}, S^2$ are also independent.

(C) The random variable

$$\frac{(n-1)\,S^2}{\sigma^2} \sim \mathrm{X}^2(n-1)$$

$$\sum_{i=1}^{r} Z_i^2 = \sum_{i=1}^{r}\left(\frac{x_i-\mu}{\sigma}\right)^2 = \sum_{i=1}^{r}\left(\frac{x_i-\bar{x}+\bar{x}-\mu}{\sigma}\right)^2$$

$$= \sum_{i=1}^{r}\left(\underbrace{\frac{x_i-\bar{x}}{\sigma}}_{A} + \underbrace{\frac{\bar{x}-\mu}{\sigma}}_{B}\right)^2$$

$$= \sum_{i=1}^{r}(A+B)^2$$

$$= \sum_{i=1}^{r}(A^2+2AB+B^2)$$

$$= \sum_{i=1}^{r}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 + 2\sum_{i=1}^{r}\left(\frac{[x_i-\bar{x}][\bar{x}-\mu]}{\sigma^2}\right) + \sum_{i=1}^{r}\left(\frac{\bar{x}-\mu}{\sigma}\right)^2$$

$$= \sum_{i=1}^{r}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 + 2\left(\frac{\bar{x}-\mu}{\sigma^2}\right)\underbrace{\sum_{i=1}^{r}(x_i-\bar{x})}_{=0} = r\left(\frac{\bar{x}-\mu}{\sigma}\right)^2$$

$$= \sum_{i=1}^{r}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 + r\left(\frac{\bar{x}-\mu}{\sigma}\right)^2 \longrightarrow \text{substitute } r \text{ for } n$$

$$= \sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 + n\left(\frac{\bar{x}-\mu}{\sigma}\right)^2$$

$$= \sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 + \left(\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}\right)^2$$

$$\underbrace{\sum_{i=1}^{n} Z_i^2}_{\chi^2(n)} = \sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 + \underbrace{Z^2}_{\chi^2(1)}$$

$$\equiv u+v+w^2$$

Since $\sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 = \sum_{i=1}^{n} Z_i^2 - Z^2$

let $u \sim \chi^2(n)$, $w \sim N(0,1) \Rightarrow w^2 \sim \chi^2(1)$, $v = \sum_{i=1}^{n}\left(\frac{x_i-\bar{x}}{\sigma}\right)^2$

$\Leftrightarrow v = u - w^2$

Thm 3.6.1 (B)

Since $\bar{x}$ and s^2 are ind. RV, and we've rearranged V into:

$$V = \sum\left(\frac{x_i-\bar{x}}{\sigma}\right)^2 = \sum\frac{(x_i-\bar{x})^2}{\sigma^2} = \frac{(n-1)s^2}{\sigma^2}$$

Recall: $s^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2$
$\Rightarrow \sum(x_i-\bar{x})^2 = (n-1)s^2$

if $v = u - w^2 \Leftrightarrow u = v + w^2$

$$M_u(t) = E(e^{tu})$$
$$= E(e^{t(v+w^2)})$$
$$= E(e^{tv})\cdot E(e^{tw^2})$$
$$= M_v(t)\cdot M_{w^2}(t)$$

$$(1-2t)^{-\frac{n}{2}} = M_v(t)\cdot(1-2t)^{-\frac{1}{2}}$$

$$M_v(t) = (1-2t)^{-\frac{n}{2}}\cdot(1-2t)^{\frac{1}{2}}$$

$$M_v(t) = (1-2t)^{-\left(\frac{n-1}{2}\right)}$$

$$\Longrightarrow v \sim \chi^2(n-1)$$

$$\Longrightarrow \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$$

$$\Longrightarrow \frac{(n-1)S_x^2}{\sigma_x^2} \sim \chi^2(n-1), \quad \frac{(m-1)S_y^2}{(m-1)} \sim \chi^2(m-1)$$

$$\Longrightarrow \frac{(n-1)S_x^2 + (m-1)S_y^2}{\sigma^2} = \frac{[(n+m-2)S_p^2]}{\sigma^2} \sim \chi^2([n+m]-2)$$

where $\sigma^2 = \sigma_x^2 = \sigma_y^2$

By Theorem 3.6.1 (B), it follows that $\widehat{\Delta} = \bar{x} - \bar{y}$ and s_p^2 are also independent, then going back to our introduction of the random variable T, the t-distribution

$$T = \frac{W}{\sqrt{V/r}}, \text{ where } W \sim N(0,1), V \sim \chi^2(r) \equiv \chi^2(n)$$

$$\frac{\dfrac{\hat{\Delta} - \Delta}{\sigma\sqrt{\frac{1}{n} + \frac{1}{m}}}}{\sqrt{\dfrac{\underbrace{(n+m-2)\cdot s_p^2}_{V}}{\underbrace{\frac{\sigma^2}{n+m-2}}_{r}}}} = \frac{\hat{\Delta} - \Delta}{\sqrt{\frac{s_p^2}{n} + \frac{s_p^2}{m}}} \sim t \text{ with } (n+m)-2 \text{ df}$$

and logically, the $100(1-\alpha)\%$ CI for a small sample size and assuming unequal variance

$$\overset{\text{est}}{\hat{\Delta}} \pm t_{\frac{\alpha}{2}, \overset{\text{df}}{n+m-2}} \cdot \sqrt{\frac{s_p^2}{n} + \frac{s_p^2}{m}}$$

Example 4.6: Let $X_1, X_2, \ldots, X_9 \overset{iid}{\sim} N(\mu, \sigma)$ [exercise 4.2.10 From Hogg]

A) if σ is known, find the length of a 95% for μ

if this instead is based on the RV

$$\frac{\bar{x} - \mu}{\sigma/\sqrt{9}} \equiv \frac{\sqrt{9}(\bar{x} - \mu)}{\sigma} \sim N(0,1)$$

Since σ is known, stick with the z-version

$$\begin{aligned}
1-\alpha &= Pr(\theta \in [L, U]) \\
&= Pr(L < \theta < U) \\
&= Pr(-z_{\frac{\alpha}{2}} < \theta < z_{\alpha/2}) \\
&= Pr\left(-z_{\frac{\alpha}{2}} < \frac{\bar{x} - \mu}{\sigma/\sqrt{9}} < z_{\frac{\alpha}{2}}\right) \\
&= Pr\left(\bar{x} - z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{9}} < \mu < \bar{x} + z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{9}}\right)
\end{aligned}$$

Thus for a 95% CI for μ, $z_{\alpha/2} = z_{0.025} = 1.96$

$\Rightarrow$ length = upper - lower (width of CI)

$$\begin{aligned}
&= \left(\bar{x} + z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{9}}\right) - \left(\bar{x} - z_{\frac{\alpha}{2}} \cdot \frac{\sigma}{\sqrt{9}}\right) \\
&= 2(1.96) \cdot \frac{\sigma}{\sqrt{9}} \\
&= 1.31 \cdot \sigma
\end{aligned}$$

B) if σ is unknown, find the expected length based on $\sqrt{9}\left(\frac{\bar{x}-\mu}{s}\right)$

$E(U-L)=?$

Hint: $E(S)=\frac{\sigma}{\sqrt{n-1}}E\left(\sqrt{\frac{(n-1)s^2}{\sigma^2}}\right)$

$$=\frac{\sigma}{\sqrt{8}}E\left(\underbrace{\sqrt{\frac{n-1}{\sigma^2}s^2}}_{V\equiv\sim\chi^2(n-1)}\right)$$ From prev. problem

$$=\frac{\sigma}{\sqrt{8}}E\left(\left[\frac{n-1}{\sigma^2}s^2\right]^{1/2}\right)$$

$$=\frac{\sigma}{\sqrt{8}}E\left(V^{\frac{1}{2}}\right)$$ where $V\sim\chi^2(8)\equiv\Gamma(\alpha=4,\beta=2)$

$$E(V^{1/2})=\int v^{\frac{1}{2}}\cdot\frac{1}{\Gamma(4)2^4}v^{4-1}e^{-\frac{v}{2}}dv$$

Recall: Thm 3.3.1 that defines any k^{th} moment for $\chi^2(r)$

$$E(v^k)=\frac{2^k\Gamma(\frac{r}{2}+k)}{\Gamma(r/2)}$$ $\alpha=\frac{r}{2}\Rightarrow\alpha=\frac{8}{4}=2$

$$E(v^{\frac{1}{2}})=\frac{2^{\frac{1}{2}}\Gamma(4+\frac{1}{2})}{\Gamma(4)}$$

$$=\frac{\sqrt{2}(3.5\cdot2.5\cdot1.5)\Gamma(0.5)}{3\cdot2\cdot1}$$

$\Gamma(\frac{1}{2})=\sqrt{\pi}$

$$=\frac{\sqrt{2}(3.5\cdot2.5\cdot1.5)\sqrt{\pi}}{3\cdot2\cdot1}$$

$$E(S)=\frac{\sigma}{\sqrt{8}}\left[\frac{\sqrt{2}(3.5\cdot2.5\cdot1.5)\sqrt{\pi}}{3\cdot2\cdot1}\right]$$

$$E(U-L)=2\cdot t_{0.025,8}\cdot\frac{E(S)}{\sqrt{9}}$$

$$=2\cdot2.306\cdot\frac{0.969\cdot\sigma}{3}=1.49\sigma$$

$U-L=1.31\sigma$ when σ is known, vs. 1.49σ where σ is unknown and relying on S

Not surprising, as $n\to\infty$ $\lim\frac{\sigma}{\sqrt{n-1}}E(v^{\frac{1}{2}})$ shows an interesting behaviour

$$\lim_{n\to\infty}\frac{\sqrt{2}\,\Gamma\left(\left[\frac{n-1}{2}\right]+\frac{1}{2}\right)}{\sqrt{n-1}\cdot\Gamma\left(\left[\frac{n-1}{2}\right]\right)}=1$$

$\Rightarrow\lim_{n\to\infty}E(S)=\lim_{n\to\infty}\frac{\sigma}{\sqrt{n-1}}E(v^{\frac{1}{2}})=\sigma$

$\Rightarrow E(S)$ is asymptotically unbiased just like $E(S^2)\approx\sigma^2$ as $n\uparrow$

Order Statistics

Previously, we talked about the formula for the sample mean (and how it has been a formula we just accepted) and how we actually generated this formula based on the MLE. Similarly, we've known the formula for the median for a while and although most of us have accepted the formula (really, an algorithm), let's talk about how we got it.

Let $X_1, X_2, \dots, X_n$ denote a random sample from a continuous random variable with some PDF $f(x)$ and support (a, b), where $-\infty < a < b < \infty$. Let Y_1 be the smallest of all X_i, and Y_2 be the next smallest X_i in terms of order of magnitude, and Y_n be the largest. That is,

$$Y_1 < Y_2 < \cdots < Y_n$$

Alternative Notation:

$$X_{(1)} < X_{(2)} < \cdots < X_{(n)}$$

We call Y_i (or $X_{(i)}$) the $\boldsymbol{i^{th}}$ **order statistic** of the random sample $X_1, X_2, \dots, X_n$. The joint PDF of $Y_1, Y_2, \dots, Y_n$ is given by the following theorem.

Theorem 4.4.1: Let $Y_1 < Y_2 < \cdots < Y_n$ denote the n order statistics based on the random sample $X_1, X_2, \dots, X_n$ from a continuous random variable with PDF $f(x)$, the joint PDF of $Y_1, Y_2, \dots, Y_n$ is

$$g(y_1, y_2, \dots, y_n) = n!\, f(y_1) \cdot f(y_2) \cdot \dots \cdot f(y_n)$$

where $a < y_1 < \cdots < y_n < b$.

Question: Why are ordered statistics important?

Answer: it is the basis and foundation of nonparametric statistics

Example 4.7: Let $X \sim$ continuous random variable where,

$$F(x) = P(X \le x) = \int_a^x f(x)\ dx, \qquad a < x < b$$

if $x \le a \Rightarrow F(x) = 0$
if $x \ge b \Rightarrow F(x) = 1$

Let $c \in \mathcal{S} \ni F(\overset{c}{x}) = 1/2$, then let X_1, X_2, X_3 denote a random, independent sample of X, and Y_1, Y_2, Y_3 denote their ordered statistic, then:

$$g(y_1, y_2, y_3) = \begin{cases} 6 \cdot f(y_1) \cdot f(y_2) \cdot f(y_3), & a < y_1 < y_2 < y_3 < b \\ 0, & \text{e/w} \end{cases}$$

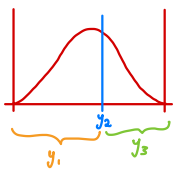
As such, since these are ordered statistics, we have to be cautious about the upper and lower bounds

The PDF of y_2

$$h(y_2) = \int_{y_2}^{b}\int_{a}^{y_2} g(y_1, y_2, y_3)\, dy_1\, dy_3$$

$$= \int_{y_2}^{b}\int_{a}^{y_2} 6 f(y_1) f(y_2) f(y_3)\, dy_1\, dy_3$$

$$= 6 f(y_2) \int_{y_2}^{b}\int_{a}^{y_2} f(y_1) f(y_3)\, dy_1\, dy_3$$



$$= 6 f(y_2) \left[\int_{y_2}^{b} f(y_3) \int_{a}^{y_2} f(y_1)\, dy_1\, dy_3 \right]$$

$$= 6 f(y_2) \underbrace{\left[1 - F(y_2)\right]}_{\Pr(Y_3 > y_2)} \underbrace{\left[F(y_2)\right]}_{\Pr(Y_1 \le y_2)}$$

To find c, then $\Pr(Y_2 \le c) = \int_a^c h(y_2)\, dy_2$

$$= \int_a^c 6 f(y_2)\left[1 - F(y_2)\right]\left[F(y_2)\right] dy_2$$

$$= 6\int_a^c F(y_2) f(y_2) - F(y_2)^2 f(y_2)\, dy_2$$

$$= 6\int_a^c \left[F(y_2) - F(y_2)^2\right] f(y_2)\, dy_2$$

keep in mind the relationship b/w the cdf and pdf for some continuous RV

$$\int F(x) f(x)\, dx = \int F(x) \frac{d}{dx} F(x)$$

$$\Rightarrow \int f(x)\, dx = \frac{F(x)^2}{2}, \text{ i.e, } F(x) \text{ is a random variable}$$

$$\frac{d}{dx}\left[\frac{F(x)}{2}\right]^2 = F(x)\frac{d}{dx}F(x) = F(x) f(x)$$

$$\Longrightarrow \Pr(Y_2 \le c) = 6\left[\frac{F(y_2)^2}{2} - \frac{1}{3}F(y_2)^3\right]\Big|_a^c = 1/2$$

thus, c = median (η), which can be estimated from the sample median (m): Y_2

based on this, the generic form for any c and prob can be found using

$$\Pr(X \le c) = \int_a^c F(x)^{\alpha - 1} \cdot f(x)\, dx = \frac{F(x)^\alpha}{\alpha}$$

$$\Pr(X > c) = \int_c^b \left[1 - F(x)\right]^{\beta - 1} f(x)\, dx = \frac{\left[1 - F(x)\right]^\beta}{\beta}$$

and extending to any k^{th} order

$$g(y_k) = \int_{y_{n-1}}^{b}\int_{y_{n-2}}^{y_n} \cdots \int_{k}^{y_{k+2}}\int_{k-1}^{y_k} \cdots \int_{a}^{y_n} n!\, f(y_1) f(y_2) \cdots f(y_n)\, dy_1\, dy_2 \cdots dy_n$$

this can be simplified

$$g(y_k) = \frac{n!}{(k-1)!(n-k)!}\left[1 - F(y_k)\right]^{n-k}\left[F(y_k)\right]^{k-1} f(y_k)$$

Example 4.8: Let $Y_1 < Y_2 < \cdots < Y_n$ be the ordered statistics of a random sample $n > 4$, where $X \sim f(x) = 2x$, where $0 < X < 1$.

if we were to express the pdf of y_3

$F(y_3) = y_3^2 \qquad f(y_3) = 2y_3$

$$g(y_3) = \frac{n!}{2!(n-3)!}\left[y_3^2\right]^2\left[1-y_3^2\right]^{n-3} \cdot 2y_3$$

let $n=4$

$$g(y_3) = \frac{4!}{2!}\left[y_3^2\right]^2\left[1-y_3^2\right]^{4-3} \cdot 2y_3 = 24\left[y_3^5 - y_3^7\right]$$

$$Pr(y_3 > \tfrac{1}{2}) = \int_{1/2}^{1} g(y_3)\,dy_3$$

$$= \int_{1/2}^{1} 24(y_3^5 - y_3^7)\,dy_3$$

$$= \frac{243}{256}$$

$$Pr(\tfrac{1}{2} < Y_1) = Pr(Y_1 > \tfrac{1}{2})$$

$$g(y_1) = \left(\frac{4!}{0!3!}\right)\left[y_1^2\right]^0\left[1-y_1^2\right]^3 \cdot 2y_1$$

$$= 8y_1 - 24y_1^3 + 24y_1^5 - 8y_1^7$$

$$Pr(Y > \tfrac{1}{2}) = \int_{1/2}^{1} 8y_1 - 24y_1^3 + 24y_1^5 - 8y_1^7\,dy_1 = \frac{81}{256} \approx 32\%$$

This question was "somewhat" intuitive because of the shape of the distribution, but your inference may have been a little off; however

if $X \sim Uni(0,1) \Rightarrow f(x) = \begin{cases} 1, & 0<x<1 \\ 0, & e/w \end{cases}$

$$\Rightarrow g(y_1) = 4 - 12y_1 + 12y_1^2 - 4y_1^3$$

$$\Rightarrow Pr(Y_1 > \tfrac{1}{2}) = 0.0625$$

As mentioned, there are multiple ways to approach <u>ANY</u> problem

$$Pr(\min\{X_i\} > \tfrac{1}{2}) = Pr(X_1 > \tfrac{1}{2}, X_2 > \tfrac{1}{2}, \ldots, X_n > \tfrac{1}{2}) \longrightarrow \text{Also works with } Y \text{ instead of } X$$

$$= Pr(X_1 > \tfrac{1}{2}) \cdot Pr(X_2 > \tfrac{1}{2}) \cdot \ldots \cdot Pr(X_n > \tfrac{1}{2})$$

$$= Pr(X_i > \tfrac{1}{2})^n \quad \forall\, i \in (1, 2, \ldots, n)$$

thus if $n=4$: $Pr(X_i > \tfrac{1}{2})^4$

$$= \left[\int_{1/2}^{1} 2x_i\,dx_i\right]^4$$

$$= \frac{81}{256} = 31.64\%$$

Let's now consider $X \sim$ continuous random variable and let some p be defined such that $0 < p < 1$, and the pth quartile of X be defined as $\xi = F^{-1}(p)$ (ξ is pronounced ksee). For example, $\xi_{0.5}$= median of X is the quantile of 0.5. Let $X_1, X_2, \dots, X_n$ be our random sample, $Y_1, Y_2, \dots, Y_n$ be our ordered statistics and $k = \max(\mathbb{Z}^+ \leq [p(n+1)])$

$$\Pr(Y \leq Y_k) = F(Y_k)$$

and

$$E(F(Y_k)) = \int F(y_k)\, g(y_k) \cdot dy_k$$

Let $z = F(y_k)$

$$\Rightarrow dz = \frac{d}{dy_k} F(y_k) = f(y_k)\, dy_k$$

$$\Rightarrow f(y_k)^{-1}\, dz = dy_k$$

$$g(y_k) = \frac{n!}{(k-1)!(n-k)!} F(y_k)^{k-1} [1 - F(y_k)]^{n-k} f(y_k)$$

$$= \frac{n!}{(k-1)!(n-k)!} z^{k-1} [1-z]^{n-k} f(y_k)$$

$$\Rightarrow E(F(y_k)) = E(z) = \int z \cdot \frac{n!}{(k-1)!(n-k)!} z^{k-1} (1-z)^{n-k} f(y_k)\, dy_k$$

$$= \frac{n!}{(k-1)!(n-k)!} \int \underbrace{z^k (1-z)^{n-k}}_{\to f(z) \text{ where } z \sim \beta \text{ which is related to the } \Gamma(\alpha,\beta) \text{ distribution}} dz$$

$$= \frac{n!}{(k-1)!(n-k)!} \cdot \frac{\Gamma(k+1)\Gamma(n-k+1)}{\Gamma(n+2)}$$

$$= \frac{n!}{(k-1)!(n-k)!} \cdot \frac{k!\,(n-k)!}{(n+1)!}$$

$$= \frac{k}{n+1}$$

Therefore,

$$E\big(P(Y \le Y_k)\big) = E\big(F(Y_k)\big) = \frac{k}{n+1} = \frac{\max(\mathbb{Z}^+ \le [p(n+1)])}{n+1}$$

Thus, Y_k will be our estimate of the quartile ξ_p, hence Y_k is the pth sample quartile, also known as the $100p^{th}$ percentile of the sample

Example: Let $n = 11$, then for $k = \max(\mathbb{Z}^+ \le [p(n+1)])$, if $p = 0.5$ (the median), then:

$k = \max\{\mathbb{Z}^+ \le 0.5(12)\} = 6$ → 6th ord. stat. is median

Thus, $Y_k = Y_6 = \text{median}$

$$\underbrace{Y_1 \le Y_2 \le Y_3 \le Y_4 \le Y_5}_{5} \le \overset{|}{\underset{|}{Y_6}} \le \underbrace{Y_7 \le Y_8 \le Y_9 \le Y_{10} \le Y_{11}}_{5}$$

However, when n is even, we ran into a small "issue"

let $n=10$, then $k = \max\{\mathbb{Z}^+ \le 0.5(11)\}$

$= \max\{\mathbb{Z}^+ \le 5.5\} = 5$

but this "contradicts" our classical defn from STAT231

thus if $\frac{(n+1)}{p} \notin \mathbb{Z}^+$

then the pth quantile is

$(1-r)y_k + r\,y_{(k+1)}$

where r is a proper fraction using a weighted average

So, for $n=10, k=5$

$(1-r)y_5 + r y_6$

$(Y_1, Y_2, \dots, Y_5)$ vs. $(Y_6, \dots, Y_{10})$

and the weight could be $r = \frac{1}{2}$

$\Rightarrow (1-\frac{1}{2})y_5 + (\frac{1}{2})y_6$

$\Rightarrow \frac{y_5 + y_6}{2}$ → classic formula for median

$Q_1 = 1^{st}$ quartile $= Pr(X \le q_1) = 0.25 = Y_{0.25(n+1)}$

$Q_2 = 2^{nd}$ quartile $= Pr(X \le q_2) = 0.50 = Y_{0.50(n+1)}$

$Q_3 = 3^{rd}$ quartile $= Pr(X \le q_3) = 0.75 = Y_{0.75(n+1)}$

Example 4.9: For a random sample of $n = 15$ from some random variable X, the following observations were recorded:

56 [1]	70 [2]	89 [3]	94 [4]	96 [5]
101 [6]	102 [7]	102 [8]	102 [9]	105 [10]
106 [11]	108 [12]	110 [13]	113 [14]	116 [15]

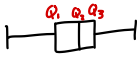
$Q_1 = Y_{0.25(n+1)} = Y_{0.25(16)} = Y_4 = 94$

$Q_2 = Y_{0.50(n+1)} = Y_{0.50(16)} = Y_8 = 102$

$Q_3 = Y_{0.75(n+1)} = Y_{0.75(16)} = Y_{12} = 108$

$$R_{(\text{default})} = \begin{cases} Q_1 = 95 \\ Q_2 = 102 \\ Q_3 = 107 \end{cases}$$

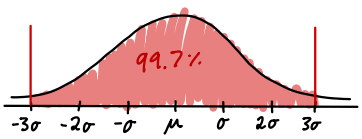
unfortunately, the methods for determining the weight r can vary heavily, which impact "where" the quantiles lie.



$IQR = Q_3 - Q_1$

$LL = Q_1 - 1.5 \cdot IQR$

$UL = Q_3 + 1.5 \cdot IQR$

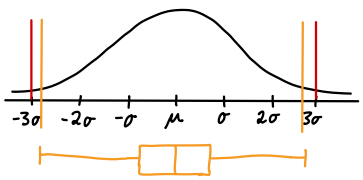


$Q_3 : P(X < q_3) = 0.75 \Rightarrow q_3 = \Phi^{-1}(0.75) \approx 0.68 \rightarrow 0.6745$

$Q_1 : P(X < q_1) = 0.25 \Rightarrow q_1 = \Phi^{-1}(0.25) \approx -0.68$

$\hookrightarrow -0.6745$

$\Rightarrow IQR = 1.36 \rightarrow 1.349$

$LL : Q_1 - 1.5 \cdot IQR = -0.6745 - 1.5(1.349) = -2.698$

$UL : Q_3 - 1.5 \cdot IQR = 0.6745 + 1.5(1.349) = 2.698$